MIL-C-13864A(EA)
AMENDMENT 6
16 August 1988
SUPERSEDING
AMENDMENT 5
10 February 1988

MILITARY SPECIFICATION

CANISTER, CHEMICAL-BIOLOGICAL MASK: M10A1

This amendment forms a part of MIL-C-13864A(EA), dated 9 February 1979, and is approved for use within the Chemical Research, Development and Engineering Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, Under "SPECIFICATIONS", delete the following documents and their titles in their entirety:

"QQ-S-781
PPP-F-320
PPP-B-621
PPP-B-636
MIL-STD-129
MIL-STD-1188"

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2.1:

Under "STANDARDS" "MILITARY", add: "MIL-STD-1168 - Ammunition Lot Numbering".

Following "DRAWINGS", change "U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER" to "U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND" and change "CHEMICAL SYSTEMS LABORATORY" to "CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER".

Under "DRAWINGS", add "136-45-842 - Tester, Canister, Shaker, Q261."

AMSC N/A FSC 4240

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Following the list of "DRAWINGS", add the following:

"PUBLICATIONS

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

SPECIAL PACKAGING INSTRUCTIONS

SPI P5-3-547 - Canister, Chemical-Biological Mask: M10A1"

2.2: Add the following new paragraphs:

"2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM STANDARDS

D 2867 - Moisture in Activated Carbon

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)"

3.3, delete in its entirety.

3.4, delete in its entirety and substitute the following:

"3.4 Filling. The chemical container shall be filled with not less than 330 milliliters of impregnated charcoal when tested as specified in 4.4.4.1. The hopper shall be gravity or hand filled from the charcoal drum. The chemical container shall be free-fall sifter filled. Updraft air flow may be used to remove fines from carbon during the free-fall filling process. The bottom shall be securely attached to the container body as shown on Drawing 5-3-549. The filter material on the finished chemical container shall form a firm covering over the screening material, free from cuts, wrinkles, tears, or other surface irregularities and shall be securely crimped."

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3.5: Delete in its entirety and substitute the following:

"3.5 Charcoal moisture content. The average moisture content of the completed canisters shall not exceed 4.0 percent by weight and no single canister shall exceed 5.0 percent by weight when tested as specified in 4.4.4.1."

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3.9:

Delete "(see 4.3.3 and 4.4.5)" and substitute "(where k = 1.8 for preproduction and k = 2.219 for regular production)".

Line 2: After "(see 6.5)" add "(computed gas life = X - (k)(s))".

Add new paragraph 3.14 as follows:

"3.14 Carbon dust emission. The dust emission shall be equal to or less than 5 milligrams (mg) when calculated by the formula AVERAGE VALUE + 2.83 (STANDARD DEVIATION) and the dust emission from any individual canister shall not exceed 5 mg when tested as specified in 4.4.4.1. After rough handling (4.4.4.1) the dust emission shall be equal to or less than 10 mg when calculated by the formula: AVERAGE VALUE + 2.83 (STANDARD DEVIATION) and the dust emission from any individual canister shall not exceed 10 mg."

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4.3.1, in line 3, change "144 canisters" to "204 packaged canisters".

4.3.2.1: Delete in its entirety and substitute the following:

"4.3.2.1 For examination and non-destructive tests. Each canister from the preproduction lot shall be examined in accordance with 4.4.3.4(a), (b) and (c) and all applicable drawing and specification requirements. Packaging shall be examined in accordance with 4.4.3.4(d) and tested as specified in 4.4.4.2. AQLs specified in 4.4.3.4 shall not apply."

4.3.2.2, TABLE II, add the following:

Test	Before rough handling	After rough handling
"Carbon dust emission	50	50"

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4.3.3: Delete in its entirety and substitute the following:

"4.3.3 Acceptance/rejection criteria. The preproduction sample shall meet criteria specified in section 3, when inspected and tested as specified in 4.3.2.1 and 4.3.2.2 to be acceptable. No regular production lot shall be accepted prior to approval of a preproduction lot. The contractor shall obtain written approval from the contracting officer before proceeding with regular production."

4.4.2.1, in line 2, change "packing canisters" to "packed canisters".

4.4.2.2, TABLE III, add the following:

Test	Before rough handling	After rough handling
"Carbon dust emission	50	50"

4.4.3.1, in line 3, change "packing canisters" to "packed canisters".

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4.4.3.2, in line 2, change "TABLE II" to "TABLE III".

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4.4.3.4(d), delete in its entirety and substitute the following:

"(d) Packaging inspection (SPI P5-3-547).

Categories		Defects
Critical:		None defined
Major:	AQL 0.25 percent defective	
101		Unit pack assembly incorrect
102		Unit pack assembly not supplied with tear strip
103		Corrosion resistance key not furnished and/or not attached to end of each can lid as specified
	AQL 2.5 percent defective	
104		Packing not as specified
105		Unit packing and packing marking missing, incorrect or illegible"

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4.4.4.1, Under Table IV, Test methods, make the following changes:

Delete "Moisture Teller" and substitute "ASTM D 2867 (g)".

Delete "Chemical Container Leakage (3.3) (d)".

Add "Carbon dust emission (3.14) Tester, Canister, Shaker, Q261, (136-45-842) (f)."

4.4.4.1(b), make the following changes:

Under "Relative Humidity", change:

FROM:	15%	TO:	15% Max
	80%		80% ± 3%
	15%		15% Max

Under "Rate of Airflow", change:

FROM:	32 liters/min[1]	TO:	32 ± 1.0 liters/min[1]
	50 liters/min[2]	TO:	50 ± 1.6 liters/min[2]
	50 liters/min[1]	TO:	50 ± 1.6 liters/min[1]

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4.4.4.1 "(d)": Delete in its entirety.

4.4.4.1, following subparagraph "(e)" add new subparagraphs "(f)" and "(g)" as follows:

"(f) The Q261 shall be used to accelerate dust emission in the canister. While the canister is being shaken, the dust shall be collected on a high efficiency filter which collects particles 0.3 microns or larger by drawing air through the canister at a constant flow rate of 50 ± 5 L/min for 20 minutes. The dust emission shall be determined gravimetrically, using a balance sensitive to 0.1 mg. The collection filter shall be either moisture-proof or shall be dessicated to insure minimal moisture absorption.

(g) Exceptions: The oven-drying method shall apply. Samples for moisture content shall be 10 to 12 g."

4.4.4.2, in title, change "Packing . . ." to "Unit pack . . .".

4.4.5, sentence 1, delete in its entirety and substitute the following:

"If any canister fails volume of filling, charcoal moisture content, carbon dust emission, or the DOP before rough handling test the lot represented shall

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be rejected. If any fails carbon dust emission after rough handling the lot shall be rejected."

Section 5, delete in its entirety and substitute the following:

"5. PACKAGING

5.1 Packaging. Packaging shall be as specified in SPI P5-3-547."

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6.1:

In subparagraph "(b)" change "(see 5.2)" to "(SPI P5-3-547)".

Change subparagraph "(c)" to "(d)" and add new subparagraph "(c)" as "(c) Palletization required (see SPI P5-3-547)".

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6.3, add the following:

"The specified resistances are to be measured at a flow rate of 85 liters per minute at 25°C and 760 millimeters of Mercury as stated in 4.4.4.1(a)."

Change: "and with filter is should" to "and with filter should".

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A954

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